

DATA COMMUNICATION LAB REPORT

Study and Plot Different MATLAB Functions Used in Data
Communication

Lab Report : One

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Aim

To study and plot different types of signals and functions used in Data Communication and Signal Processing using MATLAB.

Objectives

- To understand the concept of basic signals and functions.
- To study the behavior of Unit Step, Impulse, and Ramp functions.
- To plot Exponential Increasing and Exponential Decreasing functions using MATLAB.
- To understand the use of the `plot3()` function for 3D graphical representation.
- To gain practical knowledge of MATLAB programming for signal analysis and visualization.

Theory

1. Unit Step Function

A Unit Step Function is a basic signal used in signal processing and data communication. It changes its value from 0 to 1 at time $t = 0$. It is commonly represented by $u(t)$. This function is used to represent signals that start at a particular instant of time.

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

It is also known as the Heaviside Function.

```
t = -5 : 0.01 : 5;
unit_step = t >= 0;

figure(1);
plot(t, unit_step);
xlabel('Time');
ylabel('Amplitude');
title('Unit step signal Suraj Jha');
```

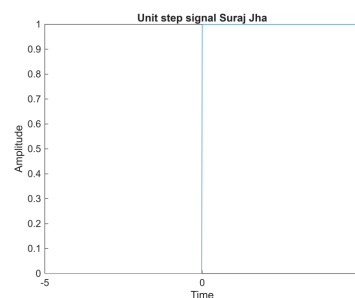


Figure 1: Unit Step Function

2. Impulse Function

An Impulse Function is a signal that has zero value at all times except at $t = 0$, where it has a very high amplitude. It is represented by $\delta(t)$. This function is mainly used in communication systems and signal analysis to study system responses.

The area under the impulse function is always equal to 1.

```
impulse = t == 0;
figure(2);
plot(t, impulse);
xlabel('Time');
ylabel('Amplitude');
title('Impulse signal Suraj Jha');
```

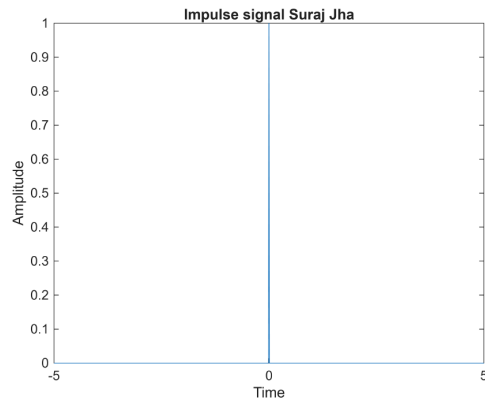


Figure 2: Impulse Function

3. Ramp Function

A Ramp Function is a signal whose amplitude increases linearly with time. It is zero for negative values of time and increases continuously for positive values of time.

$$r(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases}$$

Ramp functions are widely used in control systems and engineering applications.

```
ramp = unit_step.*t;
figure(3);
plot(t, ramp);
xlabel('Time');
ylabel('Amplitude');
title('Ramp Signal signal Suraj Jha');
```

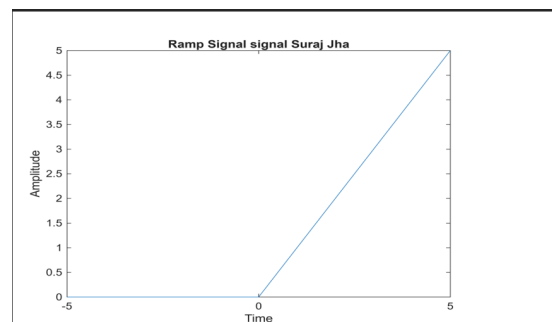


Figure 3: Ramp Function

4. Exponential Increasing Function

An Exponential Increasing Function is a signal whose amplitude increases exponentially with time.

$$x(t) = e^t$$

This function grows rapidly as time increases and is used in mathematical modeling, communication systems, and signal analysis.

```
exp_inc = 4*exp(2*t);  
figure(4);  
plot(t,exp_inc);  
xlabel('Time');  
ylabel('Amplitude');  
title('Exponential inc Suraj Jha');
```

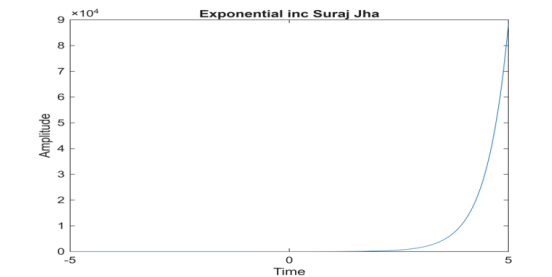


Figure 4: Exponential Increasing Function

5. Exponential Decreasing Function

An Exponential Decreasing Function is a signal whose amplitude decreases exponentially with time.

$$x(t) = e^{-t}$$

This function is commonly used to represent signal attenuation, decay processes, and discharge curves in communication systems.

```
exp_dec = 4*exp(-2*t);  
figure(4);  
plot(t,exp_dec);  
xlabel('Time');  
ylabel('Amplitude');  
title('Exponential dec Suraj Jha');
```

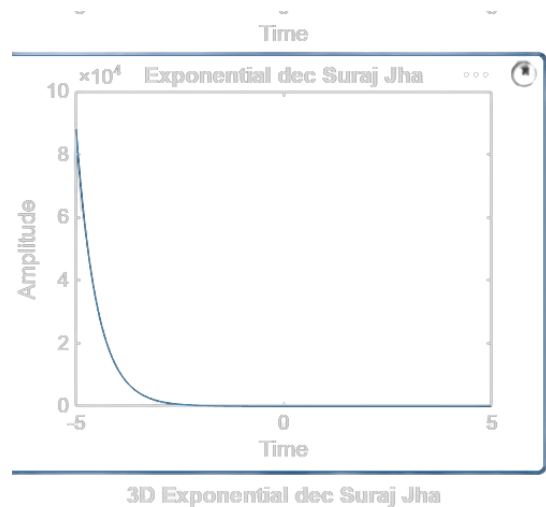


Figure 5: Exponential Decreasing Function

6. plot3 Function in MATLAB

The `plot3()` function is used to create three-dimensional plots in MATLAB. It helps visualize data in X, Y, and Z axes simultaneously.

Syntax:

$$\text{plot3}(x, y, z)$$

It is widely used in engineering, signal processing, and scientific visualization.

```
figure(5);  
plot3(t,exp_inc,exp_dec)  
xlabel('Time');  
ylabel('Exp inc');  
zlabel('Exp dec');  
title('3D Exponential dec Suraj Jha');
```

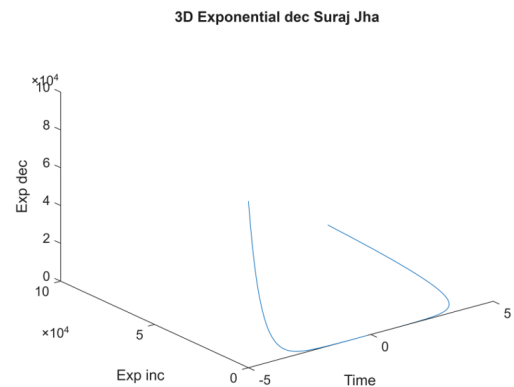


Figure 6: 3D Plot using plot3 Function

Conclusion

In this lab experiment, different MATLAB functions such as Unit Step Function, Impulse Function, Ramp Function, Exponential Increasing Function, Exponential Decreasing Function, and `plot3()` Function were studied successfully. These functions are important in Data Communication and Signal Processing for understanding signal behavior, system response, and graphical visualization. Through this experiment, practical knowledge of MATLAB programming and signal plotting techniques was achieved.